TEST 3 V1

1. It is known that a subspace Y of $\mathbf{R}^9$ can be spanned by 8 vectors, and that Y has a linearly independent set with 6 vectors. Then it is always true that:

A. $\dim Y < 6$

B. $\dim Y > 6$

C. $6 < \dim Y \leq 8$

D. $6 \leq \dim Y < 8$

(E.) $6 \leq \dim Y \leq 8$

F. None of the above is true.

We know

size of any l.i. set in Y $\leq \dim Y \leq$ size of any spanning set for Y

Hence $6 \leq \dim Y \leq 8$.

2. Suppose $\{u, v\}$ is a linearly **independent** set in vector space V, and that $w \in V$ is chosen so that $\{u, v, w\}$ is linearly **dependent**. Which of the following statements is **ALWAYS** true?

A. $\{u, w\}$ is linearly dependent.

B. $\{v, w\}$ is linearly dependent.

C. $\{v, u\}$ is linearly dependent.

D. $u \in \text{span}\{v, w\}$.

E. $v \in \text{span}\{u, w\}$.

(F.) $w \in \text{span}\{u, v\}$.

We know that if $\{u, v\}$ is l.i. and $w \in V$ then $\{u, v, w\}$ is l.i. $\iff w \notin \text{span}\{u, v\}$

Hence $w \in \text{span}\{u, v\}$.

3. Which of the following statements is true for the linear system (in 4 variables)?

$$\left.\begin{array}{rcrcrcl} x_1 & + & 3x_2 & & + & 6x_4 & = 0 \\ & & & x_3 & & & = 2 \end{array}\right\} *$$

A. The system has no solutions

B. $(-3s-6t, s, 2, t)$ is a solution for any values of s and t

C. $(-9, 1, 2, 1)$ is the unique (only) solution of the system

D. $(3s+6t, s, -2, t)$ is a solution for any values of s and t

E. $(-3s-6t, s, 0, t)$ is a solution for any values of s and t

F. $(3s+6t, s, 0, t)$ is a solution for any values of s and t

Rewriting (*) we see that

$$x_1 = -3x_2 - 6x_4$$
$$x_3 = 2$$

so if $x_2 = s$ and $x_4 = t$ are any real numbers,

$$x_1 = -3s - 6t$$
$$x_2 = s$$
$$x_3 = 2$$
$$x_4 = t$$

is a solution to (*)

4. Let $U = \{(x, y, z) \in \mathbf{R}^3 \mid x + 2y + z = 0\}$.

a) Find a basis for U and hence determine $\dim U$.

b) Give a complete geometric description of U.

c) Extend your basis in (a) to a basis of $\mathbf{R}^3$.

a) We see that $U = \{(-2y - z, y, z) \mid y, z \in \mathbb{R}\}$

$= \text{span}\{(-2, 1, 0), (-1, 0, 1)\}$.

Moreover, neither $(-2,1,0)$ nor $(-1,0,1)$ is a multiple of the other, so these 2 vectors are l.i.

Hence $\{(-2, 1, 0), (-1, 0, 1)\}$ is a basis for U.

Thus $\dim U = 2$.

b) U is the plane through 0 with normal $(1,2,1)$.

c) If we can find $w \in \mathbb{R}^3$ such that $\mathcal{B} = \{w, (-2, 1, 0), (-1, 0, 1)\}$ is l.i., then as $\dim \mathbb{R}^3 = 3$, $\mathcal{B}$ will be a basis of $\mathbb{R}^3$. But as $\{(-2,1,0), (-1,0,1)\}$ is l.i., we simply (by a theorem!) require $w \notin \text{span}\{(-2,1,0), (-1,0,1)\} = U$.

We choose $w = (1, 0, 0)$ as $1 + 2 \cdot 0 + 1 \cdot 0 = 1 \neq 0$.

i.e. $w \notin U$.

Hence $\{(1,0,0), (-2,0,1), (-1,0,1)\}$ is a basis of $\mathbb{R}^3$ extending the basis of U in (b).

5. Recall the vector space $\mathcal{P}_2 = \{a + bx + cx^2 \mid a, b, c \in \mathbf{R}\}$ of polynomial functions of degree at most 2, and define

$$W = \{p \in \mathcal{P}_2 \mid p(1) = 0\}.$$

a) Show that $W = \text{span}\{x - 1, x^2 - x\}$. *(Hint: recall the Factor Theorem: if p is any polynomial and $p(a) = 0$ for some $a \in \mathbf{R}$, then $p(x) = (x - a)q$ for some polynomial q of degree one less than that of p.)*

b) Explain why W is a subspace of $\mathcal{P}_2$ *without using the subspace test.*

c) Find a basis for W.

d) Find $\dim W$.

a) If $p(1) = 0$, then $p(x) = (x-1)(ax+b)$ for $a, b \in \mathbb{R}$ (by the Factor Theorem)
$= ax(x-1) + b(x-1)$

Hence $W = \{a(x^2-x) + b(x-1) \mid a, b \in \mathbb{R}\}$
$= \text{span}\{x^2-x, x-1\}$, as required

b) By (a) W is the span of a set of functions, and hence is a subspace of $\mathcal{P}_2$.

c) We claim $\{x^2-x, x-1\}$ is a basis. By (a), all that remains to be shown is that $\{x^2-x, x-1\}$ is l.i. So suppose $a, b \in \mathbb{R}$ satisfy $a(x^2-x) + b(x-1) = 0$, $\forall x \in \mathbb{R}$.

Then, $ax^2 + (b-a)x - b = 0$, $\forall x \in \mathbb{R}$

But $\{1, x, x^2\}$ is l.i., so $a = b-a = -b = 0$. Hence $a = b = 0$,

and so $\{x^2-x, x-1\}$ is l.i. and hence (by (a)) is a basis of W.

d) From (c), we see that $\dim W = 2$.

6. State whether each of the following statements is (always) true, or is (possibly) false, in the box after the statement.

- If you say the statement may be false, you must give an explicit example - with numbers!
- If you say the statement is always true, you must give a clear explanation.

a) If V is a vector space and $\{v_1, v_2, v_3\} \subset V$ is linearly dependent, then $\{v_1, v_2\}$ is also linearly dependent.

Let $V = \mathbb{R}^2$, $v_1 = (1,0)$, $v_2 = (0,1)$ and $v_3 = (0,0)$.

Then $\{v_1, v_2, v_3\}$ is l.d. but $\{v_1, v_2\}$ is l.i.

ANSWER FALSE

b) If u_1, u_2 and u_3 are non-zero vectors in a vector space V, and $U = \text{span}\{u_1, u_2, u_3\}$ then $\dim U = 3$.

Let $U = V = \mathbb{R}^2$, $u_1 = (1,0)$, $u_2 = (0,1)$ & $u_3 = (1,1)$.

Then u_1, u_2, u_3 are non zero, $\{u_1, u_2, u_3\}$ spans $\mathbb{R}^2$ (it contains the basis $\{(1,0), (0,1)\}$) but $\dim \mathbb{R}^2 = 2$.

ANSWER FALSE

c) If W and X are subspaces of $\mathbf{R}^2$, then their intersection

$$W \cap X = \{v \in \mathbf{R}^2 \mid v \in W \text{ and } v \in X\}$$

is closed under addition.

Let $u, v \in W \cap X$. Then $u \in W$ and $v \in W$, so $u+v \in W$ (W is a subspace). Similarly, $u \in X$ and $v \in X$ so $u+v \in X$ (X is a subspace). Hence $u+v \in W \cap X$.

TRUE

d) Given that $\{(1,1,1),(0,1,0)\}$ is a basis for a subspace W of $\mathbf{R}^3$, the coordinate vector of $(2,3,2)$ with respect to this basis is $(2,1)$.

Since $2(1,1,1) + 1.(0,1,0) = (2,2,2)+(0,1,0)$
$= (2,3,2)$,
$(2,1)$ is indeed the coordinate vector of $(2,3,2)$ w.r.t. the given basis of W.

TRUE

7. [Bonus] Suppose that u, v, w are non-zero vectors in $\mathbf{R}^{2014}$ such that $u \cdot v = u \cdot w = v \cdot w = 0$. Prove that $\{u, v, w\}$ is linearly independent. *(Hint: Use the definition. No 'geometric' argument - e.g. "they are not co-planar" - will suffice, and in any case is meaningless to low-dimensional beings like your instructor and marker.)*

Suppose $au + bv + cw = 0$ for some scalars a, b, c.

Taking dot products with u, then v, then w, we obtain

$$\begin{aligned} a\,u\cdot u + b\,v\cdot u + c\,w\cdot u &= 0 \\ a\,u\cdot v + b\,v\cdot v + c\,w\cdot v &= 0 \\ \text{and } a\,u\cdot w + b\,v\cdot w + c\,w\cdot w &= 0. \end{aligned}$$

These equations simplify to

$$\left.\begin{aligned} a\ \|u\|^2 &= 0 \\ b\ \|v\|^2 &= 0 \\ c\ \|w\|^2 &= 0. \end{aligned}\right\} *$$

Since u, v and w are all non-zero, $\|u\|^2 \neq 0$, $\|v\|^2 \neq 0$ and $\|w\|^2 \neq 0$. Thus $(*)$ implies $a = b = c = 0$.

Hence $\{u, v, w\}$ is l.i.